PHYS1408 Practice Exam 2-6 — Answer Key

Worked solutions

All answers were checked numerically with `_scripts/build_exam_2_6.py`. $g = 9.8$ m/s^2 throughout.

Practice Exam 2-6 rounds out the set: a flat-curve **maximum** (Question 1 is a round-**down** trap), string tension, a halfpipe with a starting speed, time to stop sliding **down**, spring speed partway up, a bullet into a block on a table (no swing), a light ball bouncing back, a disk spun about its rim, and walking **partway** in. The last column says which 3×5 notecard entry solves each question.

Q	Answer	Value	Topic	Notecard entry
1	**c**	14.7 m/s → 14	Flat curve, maximum speed	Curve (flat)
2	**b**	18.4 N	Hanging mass on a disk pulley, tension	Atwood / mass on pulley (one mass)
3	**d**	1.86 m	Halfpipe with a starting speed	Halfpipe
4	**a**	1.63 m/s^2	Atwood's machine, acceleration	Atwood / mass on pulley
5	**d**	2.90 s	Time to stop sliding down a rough incline	One block on incline (Stop down, Time)
6	**c**	5.35 m/s	Spring launch, speed partway up	Spring
7	**b**	1.88 m/s	Incline + hanging mass, block slides down	Incline + hanging mass + pulley
8	**e**	23.1 m	Projectile from a ledge (review)	Projectile
9	**a**	2.23 m/s	Completely inelastic: bullet into a block	Stick (inelastic)
10	**e**	−3.0 m/s	1-D elastic collision, light ball bounces back	Elastic
11	**d**	48 rad	Rotational kinematics, angle to stop	Rot. kinematics
12	**e**	0.54 kg · m^2	Parallel axis theorem, disk about its rim	Parallel axis (disk about rim)

Q	Answer	Value	Topic	Notecard entry
13	**a**	2.30 m/s^2	Rolling acceleration, hoop	Rolling (ring row)
14	**b**	+3.33 rad/s	Counter-rotating disk dropped on a disk	Angular momentum
15	**c**	1.33 rad/s	Person walks partway in	Angular momentum (walk in/out)

1. Flat curve, maximum speed — c) 14 m/s

Source: Ch 6 notes p. 2/7. **Notecard:** Curve (flat).

Notecard, flat curve ($\theta = 0$): friction alone supplies the centripetal force.

$$v_{max} = \sqrt{\mu_s g r}$$

Plug in:

$$v_{max} = \sqrt{(0.55)(9.8)(40)} = \sqrt{215.6} = 14.7 \text{ m/s}$$

Answer-choice trap. 14.7 m/s is a **maximum**. At 15 m/s the car slides, even though 15 is the closest number. The largest choice that works is **c) 14 m/s**.

Distractors:

- a) 4.7 forgets g: $\sqrt{\mu_s r}$.
- b) 10.4 has an extra $\frac{1}{2}$ under the root.
- d) 15 is the nearest number, but it's above the maximum.
- e) 19.8 drops μ_s: $\sqrt{gr}$.

2. Hanging mass on a disk pulley, tension — b) 18.4 N

Source: Ch 10 notes p. 4/8; 2015 Exam 2 #9. **Notecard:** Atwood / mass on pulley (one mass).

Notecard, one mass: $a = \dfrac{mg}{m + I/R^2}$. A disk has $I/R^2 = \frac{1}{2}M = 3.0$ kg.

Acceleration:

$$a = \frac{mg}{m + \frac{1}{2}M} = \frac{(5.0)(9.8)}{5.0 + 3.0} = \frac{49}{8.0} = 6.125 \text{ m/s}^2$$

Tension (notecard, $T = m(g - a)$ for the falling mass):

$$T = m(g - a) = 5.0(9.8 - 6.125) = 5.0(3.675) = 18.4 \text{ N}$$

Check from the pulley side: $T = \dfrac{I}{R^2}a = 3.0(6.125) = 18.4$ N (checks).

Distractors:

- a) 0 assumes a massless pulley (then nothing pulls back).
- c) 26.7 treats the pulley as a ring ($I/R^2 = M$).
- d) 30.6 is ma, not $m(g - a)$.
- e) 49.0 is mg, true only if the mass doesn't accelerate.

3. Halfpipe with a starting speed — d) 1.86 m

Source: Ch 7/8 notes p. 4/18–5/18; 2015 Exam 2 #1. **Notecard:** Halfpipe.

Notecard, one crossing ($n = 1$) with a starting speed:

$$h_f = h_i + \frac{v_i^2}{2g} - n\mu L$$

Plug in:

$$h_f = 2.0 + \frac{3.0^2}{2(9.8)} - (1)(0.12)(5.0)$$

$$= 2.0 + 0.459 - 0.60 = 1.86 \text{ m}$$

The starting speed is worth an extra $v_i^2/2g = 0.46$ m of height.

Distractors:

- a) 0.46 m is only the starting KE as height, $v_i^2/2g$.
- b) 1.26 counts the friction loss twice.
- c) 1.40 ignores the starting speed.
- e) 2.46 ignores friction.

4. Atwood's machine, acceleration — a) 1.63 m/s²

Source: Ch 7/8 notes p. 2/18. **Notecard:** Atwood / mass on pulley.

Notecard with a massless pulley ($I = 0$):

$$a = \frac{(m_1 - m_2)g}{m_1 + m_2 + I/R^2} = \frac{(m_1 - m_2)g}{m_1 + m_2}$$

Plug in:

$$a = \frac{(7.0 - 5.0)(9.8)}{7.0 + 5.0} = \frac{19.6}{12.0} = 1.63 \text{ m/s}^2$$

Distractors:

- b) 2.80 divides by m_1 only.
- c) 3.92 divides by m_2 only.
- d) 9.80 is free fall.
- e) 19.6 forgets to divide by the masses: $(m_1 - m_2)g$.

5. Time to stop sliding down a rough incline — d) 2.90 s

Source: Ch 7/8 notes p. 10/18; 2015 Exam 2 #4. **Notecard:** One block on incline (Stop down, Time).

It does stop, because $\mu_k \cos\theta = 0.435$ is bigger than $\sin\theta = 0.259$.

Step 1: Stopping distance (notecard, stop down):

$$\Delta x = \frac{v_i^2}{2g(\mu\cos\theta - \sin\theta)} = \frac{5.0^2}{2(9.8)(0.45 \times 0.9659 - 0.2588)} = \frac{25}{19.6(0.1758)} = 7.25 \text{ m}$$

Step 2: Time (notecard, with $v_f = 0$):

$$t = \frac{2\,\Delta x}{v_i + v_f} = \frac{2(7.25)}{5.0 + 0} = 2.90 \text{ s}$$

(Or by forces: $a = g(\mu\cos\theta - \sin\theta) = 1.72$ m/s², then $t = v_i/a = 2.90$ s.)

Distractors:

- a) 0.74 uses the sliding-**up** bracket, $(\sin\theta + \mu\cos\theta)$.
- b) 1.45 forgets the 2 in $2\Delta x$.
- c) 2.67 uses friction μmg instead of $\mu mg\cos\theta$.
- e) It does stop, since $\mu\cos\theta > \sin\theta$.

6. Spring launch, speed partway up — c) 5.35 m/s

Source: Ch 7/8 notes p. 12/18. **Notecard:** Spring.

Notecard spring equation, with all three terms:

$$\tfrac{1}{2}kx^2 = \tfrac{1}{2}mv^2 + mgh$$

Solve for v:

$$\tfrac{1}{2}mv^2 = \tfrac{1}{2}kx^2 - mgh$$

$$v = \sqrt{\frac{2\left(\tfrac{1}{2}kx^2 - mgh\right)}{m}}$$

Plug in: spring PE $= \tfrac{1}{2}(900)(0.080)^2 = 2.88$ J; gravity PE gained $= (0.15)(9.8)(0.50) = 0.735$ J.

$$v = \sqrt{\frac{2(2.88 - 0.735)}{0.15}} = \sqrt{\frac{4.29}{0.15}} = \sqrt{28.6} = 5.35 \text{ m/s}$$

Distractors:

- a) 3.13 is $\sqrt{2gh}$ (no spring at all).
- b) 3.78 drops the $\frac{1}{2}$ on the KE.
- d) 6.20 ignores the height gained (no mgh).
- e) 8.19 drops the $\frac{1}{2}$ on the spring PE.

7. Incline + hanging mass, block slides down — b) 1.88 m/s

Source: Ch 7/8 notes p. 17/18–18/18. **Notecard:** Incline + hanging mass + pulley.

Notecard with m_1 sliding **down** and a massless pulley ($I = 0$):

$$v_f = \sqrt{v_i^2 + \frac{2g\,\Delta x\,B}{m_1 + m_2}}$$

$$B = m_1 \sin\theta - m_2 - \mu m_1 \cos\theta$$

Net pull ($\sin 40^\circ = 0.6428$, $\cos 40^\circ = 0.7660$):

$$B = 12(0.6428) - 3.0 - 0.25(12)(0.7660) = 7.714 - 3.0 - 2.298 = 2.416 \text{ kg}$$

Plug in:

$$v_f = \sqrt{1.0^2 + \frac{2(9.8)(0.80)(2.416)}{12 + 3.0}}$$

$$= \sqrt{1.0 + \frac{37.88}{15}} = \sqrt{1.0 + 2.526} = \sqrt{3.526} = 1.88 \text{ m/s}$$

$B > 0$, so the block speeds up (e is wrong).

Distractors:

- a) 1.59 forgets v_i^2.
- c) 2.04 divides by m_1 only.
- d) 2.43 ignores friction.
- e) comes from using the wrong B (the m_2-falls version, which is negative).

8. Projectile from a ledge (review) — e) 23.1 m

Source: 2015 Exam 2 #2. **Notecard:** Projectile.

Notecard:

$$H = h_0 + \frac{(v_0 \sin\theta)^2}{2g}$$

Plug in ($\sin 55^\circ = 0.8192$, so $v_0 \sin\theta = 14.75$ m/s):

$$H = 12 + \frac{(14.75)^2}{2(9.8)} = 12 + \frac{217.4}{19.6} = 12 + 11.1 = 23.1 \text{ m}$$

Distractors:

- a) 5.4 uses $\cos\theta$ and forgets the ledge.
- b) 11.1 forgets to add the ledge.
- c) 12.0 is just the ledge.
- d) 17.4 uses $\cos\theta$ instead of $\sin\theta$.

9. Completely inelastic: bullet into a block — a) 2.23 m/s

Source: Ch 9 notes p. 1/5. **Notecard:** Stick (inelastic).

They stick, so momentum only (notecard, block at rest):

$$v_f = \frac{m_1 v_1 + m_2 v_2}{m_1 + m_2} = \frac{mv}{m + M}$$

Plug in:

$$v_f = \frac{(0.015)(300)}{0.015 + 2.0} = \frac{4.5}{2.015} = 2.23 \text{ m/s}$$

Distractors:

- b) 2.25 leaves the bullet's mass out of the denominator (close; keep 3 digits).
- c) 4.5 is the momentum (kg · m/s), not a speed.
- d) 25.9 uses energy conservation through the collision (never do that).
- e) 300 is the bullet's speed before.

10. 1-D elastic collision, light ball bounces back — e) −3.0 m/s

Source: Ch 9 notes p. 2/5–3/5. **Notecard:** Elastic.

Notecard, right = +: $m_1 = 0.50$, $v_{1i} = +6.0$; $m_2 = 1.5$, $v_{2i} = 0$.

$$v_{1f} = \frac{(m_1 - m_2)v_{1i} + 2m_2 v_{2i}}{m_1 + m_2} = \frac{(0.50 - 1.5)(6.0) + 0}{2.0} = \frac{-6.0}{2.0} = -3.0 \text{ m/s}$$

The light ball bounces back to the left. (The 1.5 kg ball leaves at $v_{2f} = 6.0 - 3.0 - 0 = +3.0$ m/s.)

Distractors:

- a) +6.0 is no change.
- b) +3.0 is the 1.5 kg ball's velocity.
- c) +1.5 is the stick-together answer.
- d) 0 is the equal-mass (cue ball) result, which needs equal masses.

11. Rotational kinematics, angle to stop — d) 48 rad

Source: Ch 10 notes p. 1/8. **Notecard:** Rot. kinematics.

Time isn't given or asked for, so use the equation without it:

$$\omega_f^2 = \omega_i^2 + 2\alpha\,\Delta\theta$$

Solve for $\Delta\theta$ ($\omega_f = 0$, and $\alpha = -1.5$ rad/s² because it slows):

$$0 = \omega_i^2 + 2\alpha\,\Delta\theta$$

$$\Delta\theta = \frac{-\omega_i^2}{2\alpha}$$

Plug in:

$$\Delta\theta = \frac{-(12)^2}{2(-1.5)} = \frac{144}{3.0} = 48 \text{ rad}$$

Distractors:

- a) 7.6 is the answer in revolutions, $48/2\pi$.
- b) 8.0 is the time to stop, ω_i/α, in seconds.
- c) 24 has an extra factor of 2 in the denominator.
- e) 96 forgets the 2: ω_i^2/α.

12. Parallel axis theorem, disk about its rim — e) 0.54 kg · m²

Source: Ch 10 notes p. 2/8. **Notecard:** Parallel axis (disk about rim).

Notecard: disk about a point on its rim, $I = \frac{3}{2}MR^2$. (That's $I_{cm} + md^2$ with $d = R$: $\frac{1}{2}MR^2 + MR^2$.)

$$I = \tfrac{3}{2}MR^2$$

Plug in:

$$I = \tfrac{3}{2}(4.0)(0.30)^2 = 1.5(4.0)(0.090) = 0.54 \text{ kg} \cdot \text{m}^2$$

Distractors:

- a) 0.12 uses the rod-end fraction, $\frac{1}{3}MR^2$.
- b) 0.18 is I_{cm} alone.
- c) 0.27 uses $d = R/2$.
- d) 0.36 is MR^2 alone (forgot I_{cm}).

13. Rolling acceleration, hoop — a) 2.30 m/s²

Source: Ch 10 notes p. 3/8. **Notecard:** Rolling (ring row).

Notecard, ring (hoop) row: $a = \frac{1}{2}g\sin\theta$.

Plug in:

$$a = \tfrac{1}{2}(9.8)\sin 28^\circ = \tfrac{1}{2}(9.8)(0.4695) = 2.30 \text{ m/s}^2$$

Distractors:

- b) 3.07 is a disk, $\frac{2}{3}g\sin\theta$.
- c) 3.29 is a sphere, $\frac{5}{7}g\sin\theta$.
- d) 4.33 uses $\cos\theta$.
- e) 4.60 is sliding, $g\sin\theta$.

14. Counter-rotating disk dropped on a disk — b) +3.33 rad/s

Source: Ch 11 notes p. 3/8; 2015 Exam 2 #11. **Notecard:** Angular momentum.

$L_i = L_f$, with the dropped disk's ω_2 **negative**:

$$\omega_f = \frac{I_1\omega_1 + I_2\omega_2}{I_1 + I_2}$$

Moments of inertia first:

$$I_1 = \tfrac{1}{2}(40)(0.50)^2 = 5.0 \text{ kg} \cdot \text{m}^2$$

$$I_2 = \tfrac{1}{2}(20)(0.50)^2 = 2.5 \text{ kg} \cdot \text{m}^2$$

Plug in:

$$\omega_f = \frac{(5.0)(6.0) + (2.5)(-2.0)}{5.0 + 2.5} = \frac{30 - 5.0}{7.5} = +3.33 \text{ rad/s}$$

Distractors:

- a) 2.00 treats the top disk as a ring (mr^2).
- c) 4.00 ignores the top disk's spin.
- d) 4.67 forgets the minus sign.
- e) 6.00 is no change.

15. Person walks partway in — c) 1.33 rad/s

Source: Ch 11 notes p. 7/8. **Notecard:** Angular momentum (walk in/out).

Notecard, walk in/out: $\omega_f = \dfrac{I_i}{I_f}\omega_i$. The person's $I = mr^2$ changes from $r = 3.0$ m to $r = 1.5$ m.

$$\omega_f = \frac{\frac{1}{2}MR^2 + mR^2}{\frac{1}{2}MR^2 + m\left(\frac{R}{2}\right)^2}\,\omega_i$$

Moments of inertia: disk $= \frac{1}{2}(300)(3.0)^2 = 1350$; person at the rim $= 75(3.0)^2 = 675$; person halfway $= 75(1.5)^2 = 168.75$ (all kg · m²).

Plug in:

$$\omega_f = \frac{1350 + 675}{1350 + 168.75}(1.0) = \frac{2025}{1518.75} = 1.33 \text{ rad/s}$$

The rim→center shortcut $\frac{M+2m}{M}$ only works for walking **all the way** to the center.

Distractors:

- a) 0.75 flips the ratio (walking out).
- b) 1.00 is no change.
- d) 1.50 uses the rim→center shortcut, as if they walked all the way in.
- e) 2.00 assumes halving the distance doubles ω.